ANSHUL PAL

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TECHNICAL SKILLS

Languages:	Go, C, Python, Bash / Shell, SQL (PostgreSQL), TypeScript, JavaScript
Cloud Infra & Systems:	Linux Namespaces, Cgroups v2, Container Runtimes, POSIX Syscalls, Process Management, Distributed Consensus (Raft)
Networking & Protocols:	TCP/IP Sockets, RPC, HTTP Lifecycle, SMTP (RFC 5321/5322), IPC (Pipes, File Descriptors, Signals)
Reliability & DevOps:	High Availability & Failover, Quorum Math, Infrastructure as Code (render.yaml), Docker, Make, GCC, Git, Supabase

TECHNICAL PROJECTS

Miniraft – Fault-Tolerant Distributed Consensus & Lock Service github.com/anshul1729-in/miniraft
Go, TCP RPC, Distributed Consensus, High Availability *Oct 2026*

- Architected a fault-tolerant distributed Key-Value store and Lock Manager in Go implementing the Raft Consensus Protocol from scratch over raw TCP RPCs.
- Engineered automated leader election with randomized election timers (300–600ms) and strict quorum math (⌊N/2⌋ + 1), eliminating split-brain risks during network partitions.
- Implemented two-phase commit log replication (`AppendEntries`), validating 100% data durability and surviving abrupt `kill -9` leader failure in ~400ms with zero committed log loss.
- Formulated idempotent state mutation handling via client-generated `req_id` deduplication tracking, preventing redundant execution during leader transitions and network retries.

Minicontainer – Linux Container Runtime from Scratch github.com/anshul1729-in/minicontainer
Go, Linux Namespaces, Cgroups v2, Syscalls, Resource Throttling *Oct 2026*

- Engineered a lightweight container runtime in Go directly on Linux kernel primitives via Go `syscall`, achieving complete process isolation without Docker or containerd.
- Isolated process trees, hostnames, mount tables, and network interfaces via `CLONE_NEWPID`, `CLONE_NEWUTS`, `CLONE_NEWNS` (`MS_PRIVATE`), and `CLONE_NEWNET` namespaces.
- Implemented secure filesystem isolation using `pivot_root` and host root detachment (`MNT_DETACH`) over Alpine Linux minirootfs with dedicated `/proc` mount.
- Enforced strict hardware quotas via Linux **Cgroups v2** (`/sys/fs/cgroup`), capping container memory at 50MB (swap disabled) and throttling CPU utilization to a 50% core quota.
- Resolved Linux PID 1 namespace re-parenting and execution lifecycle using the `/proc/self/exe` re-execution architecture.

Minishell – POSIX Systems Process Orchestrator github.com/anshul1729-in/minishell
C, Linux Syscalls, Process Management, POSIX API *Oct 2026*

- Engineered a custom POSIX-compliant shell in C orchestrating process lifecycles via low-level kernel syscalls (`fork()`, `execve()`, `waitpid()` with `WUNTRACED` and `WNOHANG`).
- Implemented an arbitrary-depth multi-stage pipeline executor (`cmd1 | cmd2 | cmd3`) by dynamically rewiring per-process File Descriptor tables using `pipe()` and `dup2()`.
- Built comprehensive I/O redirection (`<`, `>`, `2>&1`) and terminal session job control (`setpgid()`, `tcsetpgrp()`) to isolate foreground and background process groups.
- Designed asynchronous signal multiplexing for `SIGINT` (Ctrl+C), `SIGTSTP` (Ctrl+Z), and `SIGCONT`, maintaining an interactive process table for job control (`jobs`, `fg`, `bg`).

Custom Mail Server & Cloud IaC – Network Protocols & Deployment github.com/anshul1729-in/personal-email-server
Python, aiosmtpd, Node.js, Render IaC, PostgreSQL *Jul 2026*

- Engineered an asynchronous custom SMTP mail server in Python (`aiosmtpd`) processing raw RFC 5321 and RFC 5322 protocol streams over non-privileged TCP ports.
- Automated production cloud deployments using Infrastructure as Code (`render.yaml`), orchestrating containerized web backends and managed PostgreSQL databases.
- Implemented structured in-memory TTL caching layers (60,000ms) on backend services to prevent analytical database contention and eliminate thundering herd traffic.

WORK EXPERIENCE

PropertyLay Jun 2025 – Sep 2025
Freelance Software Engineer *New Delhi, India*

- Architected and delivered a responsive commercial real estate web platform using Next.js (App Router), React, and Tailwind CSS for dynamic property portfolios.
- Engineered client-side routing, modular UI component hierarchies, and performant dynamic lead capture forms integrated with input validation pipelines.
- Optimized asset delivery, metadata, and Core Web Vitals, achieving sub-second Largest Contentful Paint (LCP) and zero-layout-shift UI transitions.

EDUCATION

Amity University Expected 2027

Bachelor of Computer Applications (BCA) – Computer Science *Noida, India*

- **Relevant Coursework:** Operating Systems, Computer Networks, Distributed Systems, Database Management Systems, System Programming, Cloud Computing.